Anthony T. Pulvino

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Education

Doctor of Philosophy, Biological Sciences **2026**
Northwestern University, Evanston, IL

Master of Science, Biology (Bioinformatics and Genomics Program) **2020**
University of Oregon, Eugene, OR

Bachelor of Science, Majors: Anthropology; Biological Sciences **2018**
University of Wisconsin - Milwaukee, Milwaukee, WI

Publications

Pulvino, A.T., Quinn W.G., Amato, K.R. (2026). Functional metabolic and eco-evolutionary categories partition a multi-resolution analysis workflow, revealing unreported primate coding sequences under positive selection. *Heredity* (In-Review).

Pulvino, A.T. (2026). *evolharvester* enables collection and ordered structuring of multi-resolution HyPhy results. *Bioinformatics Advances* (In-Review).

Pulvino, A.T., Savo Sardaro, M.L., Quinn, W.G., Christopher, E.R., Koenig, S., Lucore, J., Vidal-Cruchez, O., Mallott, E.K., Kuthyar, S., Lee, W., Chitta, S., Gray, S., Williams, L.E., Curley, J.P., Liechty, E.R., Ben-Sahra, I., and Amato, K.R. (2026). Bile acid co-metabolic disruption mediates macaque-donor adiposity in a germ-free, C57BL/6 mouse model. *Microbial Genomics* (In-Review).

Bailon-Zambrano, R., Sucharov, J., Mumme-Monheit, A., Murry, M., Stenzel, A., Pulvino, A.T., Mitchell, J.M., Colborn, K.L., and Nichols, J.T. (2022). Variable paralog expression underlies phenotype variation. *ELife* 11:e79247. https://doi.org/10.7554/eLife.79247

Uppala, J. K., Sathe, L., Chakraborty, A., Bhattacharjee, S., Pulvino, A. T., & Dey, M. (2022). The cap-proximal RNA secondary structure inhibits preinitiation complex formation on HAC1 mRNA. *Journal of Biological Chemistry*, *298*(3). https://doi.org/10.1016/j.jbc.2022.101648

Mitchell, J. M., Sucharov, J., Pulvino, A. T., Brooks, E. P., Gillen, A. E., & Nichols, J. T. (2021). The alx3 gene shapes the zebrafish neurocranium by regulating frontonasal neural crest cell

differentiation timing. *Development (Cambridge, England)*, *148*(7), dev197483. https://doi.org/10.1242/dev.197483

Research Experience

Graduate Student Consultant - Interdisciplinary Assistantship **2024-Present**
Research Data Services (Data Sci, Stats, and Viz team) at Northwestern University
● Provide support for trans-disciplinary projects, supporting data analysis efforts from single cell RNA-seq to text-based analysis of archival documents from the French revolution
● Assisted with installation of software from binaries as well as more complicated virtual environments for genomics projects
● Updated materials for and lead the 'Web-scraping in R using *Rvest'* workshop in RDS's "Next Steps in R" series
● Served as member on Academic Advisory Committee for beta-testing of the Research Computing's rollout of the *ColdFront* interface for supercomputing resource management
● Contributed to analysis of twitter data collected via API during the COVID pandemic to gain insights using a variety of text-based analysis tools within custom scripts
● Assisted as logistical support during the inaugural CoDEx symposium hosted by Research Computing and Data Services teams at Northwestern

Graduate Student Consultant **2022-2023**
Research Computing Services (Genomics Compute Cluster) at Northwestern University
● Provide high-performance computing (HPC) consulting project support through ticketing system for Northwestern researchers working with 'omics-scale datasets
● Have made new software modules available through Quest for genomics researchers including Qiime2 and Cutadapt.
● Helped curate workshops for b1042/Genomics Compute Cluster users and wrote user-friendly software documentation for deploying Purdue Biocontainers on Quest
● Contributed to resource allocation projects to improve the speed and efficiency of job submissions to the GCC for researchers.

Professional Research Assistant **2020-2021**
Department of Craniofacial Biology at the University of Colorado's Anschutz Medical Campus (Nichols Lab)
● Assessed quality of genomic assemblies of *Petromyzon marinus* (the sea lamprey) for downstream single cell RNA-seq analyses
● Contributed to exploratory, bulk RNA-seq experimental design and data analysis to assess a mutant expressing a novel phenotype in a zebrafish model
● Prepared scRNA-seq data analyses to compare differential expression profiles between *Mef2ca* mutant and wild type zebrafish.
● Attended and presented at lab meetings and coordinated data exchange and communications with collaborators for newly sequenced assembly data
● Developed experimental design, conducted wet laboratory preparations from mutant zebrafish displaying incomplete penetrance

● Developed RShiny apps for co-authored, peer reviewed publication and contributed to writing and revisions, and figures included in both main manuscript and supplement

Graduate Research Assistant **2019-2020**
Department of Biology at the University of Oregon; Stowers Institute for Medical Research (Yu Lab)
● Annotated single cell populations from scRNA-seq data derived from mouse vomeronasal organ using bioinformatic techniques
● Coordinated group meetings and organized team responsibilities
● Determined neurodegenerative diseases associated with differentially expressed genes of the mouse vomeronasal organ
● Composed a finalized report of our research and findings

Teaching Experience

Teaching Assistant (ANTHRO490: Microbiome Analysis) **2022**
Anthropology Department at Northwestern University
● Assisted during lectures to clarify on introductory information related to high-performance computing, leveraging the linux CLI environment for system navigation and analysis, prepared introductory GitHub use training module
● Assisted students in with questions on in-class activities which assessed their fundamental knowledge of data analysis tools for microbiome research (qiime2, humann2)

Teaching Assistant (BIO202: Cell Biology) **2022**
Molecular Biosciences Department at Northwestern University
● Assisted students in office hours with questions on homework and previous exams which assessed their fundamental knowledge of cell biology
● Gave exam-review session lectures/discussion-based and fielded questions during scheduled course lectures alongside professor/other TAs.

Teaching Assistant (BIO203: Genetics and Evolution) **2022**
Molecular Biosciences Department at Northwestern University
● Assisted students in office hours with questions on homework and previous exams which assessed their fundamental knowledge of genetics and evolution
● Demonstrated strong interpersonal skills filling in for co-TAs and answering questions alongside the professors after each lecture
● Motivated students to actively learn during office hours and exams by answering their questions with questions, inviting critical and logical thinking about biological problems

Presentations and Posters

DHVI Seminar (Talk) **2026**
Duke Human Vaccine Institute at Duke University

Systems of host-microbe interaction inform host species metabolic differences across primates

Human Evolutionary Biology Seminar (Talk) **2026**
Human Evolutionary Biology at Harvard University
Systems of host-microbe interaction inform host species metabolic variation across primates

C-MAIKI Seminar (Talk)
C-MAIKI at University of Hawaii – Manoa **2026**
Systems of host-microbe interaction inform host species metabolic differences across primates

Pulvino, A.T., Quinn, W.G., Christopher, E.R., and Amato, K.R. **(Poster)** **2025**
Department of Anthropology at Northwestern Univeristy
"ScRNA-seq of the Mouse Vomeronasal Organ Reveals New Insights Into Innate Behaviors." Presented at "American Society of Primatologists and Midwest Primate Interest Group Meetings (2025)."

Alumni Section Conference (Talk) **2024**
Bioinformatics and Genomics Master's Program at University of Oregon

***CLP* Research Forum (Talk)**
Chemistry of Life Processes (CLP) at Northwestern University **2023**
Exploring paths between the primate microbiome and its influence on host gene regulation

***CLP* Research Forum (Talk)**
Chemistry of Life Processes (CLP) at Northwestern University **2022**
Microbiota metabolites and endogenous retrovirus interactions within a murine cecal epithelial cell context

Guest Lecturer (BIO572: Functional Genomics; Talk) **2021**
Biological Sciences Department at University of Wisconsin - Milwaukee

Biondi Thomas, Ryan Lancione, Pulvino A, Yu Ron. **(Poster)** **2020**
Department of Biology at the University of Oregon
"ScRNA-seq of the Mouse Vomeronasal Organ Reveals New Insights Into Innate Behaviors." Presented at "Genomics In Action Conference (2020)."

Klitzke N, Ghosh C, Pulvino A, Dey M. **(Poster)** **2019**
Department of Biological Sciences at the University of Wisconsin - Milwaukee
"Unfolded Protein Response Pathway Coupled to Activation of Slt2 MAP Kinase Pathway in the Budding Yeast *Saccharomyces cerevisiae.*" Presented at "11th Annual UWM Undergraduate Research Symposium."

Pulvino A, Dey M. **(Poster)** **2018**

Department of Biological Sciences at the University of Wisconsin - Milwaukee
"Increasing Blood Oxygen Carrying Capacity: Structure-Function of HIF-2alpha Protein in Review and Directions for Future Research." Presented at "10th Annual UWM Undergraduate Research Symposium."

Awards

● Research Computing and Data Services Interdisciplinary Assistantship
Northwestern University, Evanston, IL. (**2023-2026)**

● Bioinformatics and Genomics Master's Program Alumni Mentor Program Honorarium
University of Oregon, Eugene, OR. (**2023)**

● Chemistry of Life Processes T32 Training Grant Awardee
Northwestern University, Evanston, IL. (**2022-2023**)

● SURF Undergraduate Research Fellowship Awardee
University of Wisconsin - Milwaukee, Milwaukee, WI. (**2017-2018)**